SUPPORTING ADULT GUIDANCE

Inequality symbols and double-ended inequalities

Inequalities · Lesson 1. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson introduces the four **inequality symbols** and **double-ended inequalities**. The central idea is that an inequality describes a **set** of values, and that each boundary value is either included (≤, ≥) or excluded (<, >). Learners sort numbers through a rule, list the whole numbers that satisfy it, and translate everyday boundaries (at least, at most, between …) into inequalities.

Driving Question: *Who gets through the gate?*

The mathematics, in plain terms

a < b means a is to the left of b on the number line, and is equivalent to b > a. A non-strict inequality (≤, ≥) includes the boundary value; a strict one (<, >) excludes it. A double-ended inequality a < x ≤ b requires both parts to hold, so its solution set is the stretch of the number line between a and b, with each end open or closed according to its symbol.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Through the gate?

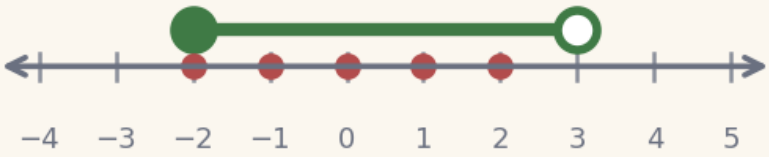
The gate's rule is $x \geq -2$. Do –2, –3 and 0 get through?

1. Test –2: is –2 ≥ –2? It is equal, and ≥ includes equal.	–2 gets through
2. Test –3: is –3 ≥ –2? –3 is to the left of –2, so it is smaller.	–3 does not
3. Test 0: is 0 ≥ –2? 0 is to the right of –2.	0 gets through

–2 and 0 get through; –3 does not. The filled circle at –2 shows that the boundary is included.

Example 2 — Listing whole numbers

List the whole numbers n with $-2 \leq n < 3$.



1. Lower boundary: ≤ includes –2.	start at –2
2. Step up one at a time, remembering 0.	–2, –1, 0, 1, 2
3. Upper boundary: < leaves 3 out.	stop before 3

–2, –1, 0, 1, 2 — five whole numbers. Filled circle at –2 (included), open circle at 3 (left out).

Meaning before procedure

Let learners decide number by number before naming any rule. The gate activity is designed so that the boundary numbers come up early and often: those are the decisions that build understanding. Ask for the reason every time, not only the decision. Negative numbers appear throughout, because comparing negatives is the most common gap; the number line is the reference to return to.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Compare two numbers on a number line and choose <, = or >; write the same fact both ways.	~8 min
Movement	Tap the whole numbers that obey a double-ended rule; check each boundary.	~10 min
Reflection	The boundary gate: sort numbers, give reasons, and watch the solution set appear.	~13 min
Creativity	Turn real-world sentences into inequalities; stretch: make your own boundary.	~9 min
Rest	Notice one boundary in your own day: ≤ or <?	~2 min
Nutrition	Intellectual: precision about limits — safety, fairness and health rules.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Thinking −7 is greater than −3 because 7 is greater than 3.
Return to the number line: 'Which is further to the right?' Temperatures help: −7 °C is colder than −3 °C.

Including a boundary that a strict inequality leaves out (or the reverse).
Ask them to test the boundary in the rule: 'Is 3 < 3 true?' Then link: < and > leave it out; ≤ and ≥ let it in.

Missing 0 when listing whole numbers between a negative and a positive boundary.
'Say every whole number from −2 up to 2 out loud.' Zero is often skipped when it is not written down.

Reading the symbol the wrong way round.
'The open side faces the greater number.' Read the statement aloud in words to check it is true.

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: ordering numbers, including negative numbers, on a number line. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says 3 gets through the gate for $-1 < x < 3$, or lists −2 for $-2 < n \leq 2$. The aim is a question that returns the reasoning to them.

- "Put 3 into the rule. Is 3 < 3 true?"
- "Which symbol is next to that boundary? What does it say about the boundary itself?"

- “Which is further right on the line: –7 or –3? So which is greater?”
- “Name one number just inside the boundary and one just outside. Do they behave as you expect?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why spend a whole lesson on symbols?

Because the decision at each boundary is where most later errors come from. Solving inequalities and drawing them on a number line (Lessons 2 and 3) depend on reading the symbols securely.

Do the numbers have to be whole numbers?

No. An inequality includes every value in its range, including decimals such as 2.5. Some questions ask only for the whole numbers (integers), and say so.

How does this connect to the diagnostic?

The Walled Garden diagnostic checked ordering negatives and whole numbers between two bounds — the two ideas this lesson relies on most.

What comes next?

Lesson 2 solves inequalities by doing the same to both sides, and discovers why multiplying or dividing by a negative number reverses the inequality.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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